The Nexus of Macroeconomic Policy Quality on Inflation: An Examination on Government Regulation and Its Effect on Consumers

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Abstract

This paper investigates the relationship between the quality of a nation's macroeconomic policy, as measured by the World Bank's CPIA Macroeconomic Management Index, and its annual inflation rate. Using a panel dataset of 75 countries from 2005 to 2016, we estimate several fixed effects models to examine this relationship while controlling for credit conditions, export volumes, and broader measures of institutional and financial development. Results indicate that while higher macroeconomic policy ratings are associated with lower inflation rates, this relationship is moderated by a country's export volume. Furthermore, macroeconomic policy quality only achieves statistical significance when accounting for domestic financial sector credit growth and broader structural factors such as social equity, debt management, and property rights protections. These findings suggest that macroeconomic policy effectiveness is deeply intertwined with a nation's financial, social, and institutional development.

Introduction

The global economic landscape has recently been shaped by a resurgence of inflation, reaching levels not seen in several decades across many developed nations within the Organization for Economic Co-operation and Development (OECD). This inflationary surge has escalated living costs, disproportionately burdening lower-income populations. Inflation's impact transcends economics, fueling civil unrest and political instability, with significant consequences for national governance. These dynamics necessitate a nuanced understanding of the underlying drivers of inflation, especially the role of macroeconomic management.

Political outcomes are increasingly sensitive to inflationary trends. In countries like the United States, inflation has emerged as a primary concern for voters, influencing electoral outcomes and reshaping policy debates. Voter dissatisfaction linked to sustained inflation, even as rates stabilize, underscores the profound political and economic implications of price instability.

Given inflation's pivotal role in economic and political arenas, this paper examines whether a country's quality of macroeconomic management, as captured by the World Bank's CPIA Macroeconomic Management Index, meaningfully affects inflation outcomes. Preliminary evidence, depicted in Figure 1, suggests a slight negative correlation between policy quality and inflation rates, motivating a deeper econometric investigation.

Literature Review

Economic theory and empirical research identify several pathways through which macroeconomic quality shapes inflation. The Phillips curve posits a relationship between output and inflation, with faster GDP growth or lower unemployment leading to upward price pressures (Phillips, 1958; Friedman, 1968). Empirical findings (Coibion & Gorodnichenko, 2015) confirm that demand-side pressures intensify inflation, potentially necessitating GDP controls.

Income levels, proxied by GDP per capita, also modulate inflation dynamics. Developed economies tend to exhibit stable, low inflation rates, whereas developing nations often experience greater volatility (Ha, Kose, & Ohnsorge, 2019). Credit growth similarly plays a vital role; rapid financial expansion can fuel demand and thus inflation (Choi & Cook, 2018).

Lastly, external trade shocks matter significantly. Export booms or trade price volatility affect domestic inflation via terms-of-trade mechanisms (Comin & Johnson, 2020). Additionally, financial sector quality and governance structures influence monetary policy effectiveness. Strong banking systems and regulatory frameworks help stabilize inflation (Mishkin, 2007), whereas corruption and weak institutions exacerbate volatility (Mujahid, 2021). Infrastructure deficiencies, by creating supply bottlenecks, can further destabilize prices (Straub, 2011).

Thus, controlling for GDP growth, income levels, credit conditions, trade shocks, financial sector depth, governance quality, and infrastructure is critical to isolating the impact of macroeconomic management on inflation. This study controls for each of these sources of bias by including statistically significant variables in the finalized regression model.

Data Description

Table 1:– Variable Descriptions

Inflation	Inflation in consumer prices (annual %)
Macro	CPIA macroeconomic management rating (1=low to 6=high)
CPIA_Avg	A composite index calculated as the mean of CPIA ratings for economic management, social inclusion, financial sector development, and property rights
Export	Export Volume Index (Standardized in comparison to 2,000)
Credit	Domestic Credit provided to financial sector (% of GDP)

This research employs panel data collected at the country-year level between 2005 and 2016 to examine the relationship between macroeconomic policy quality and inflation rates. The dataset contains a total of 922 observations, covering 75 countries across a diverse range of economic, institutional, and developmental contexts. Each observation corresponds to a unique country-year pair, ensuring temporal and cross-sectional variability essential for robust fixed-effects analysis.

The primary dataset is the World Bank's Country Policy and Institutional Assessment (CPIA) database. The CPIA provides ratings across several dimensions of governance and economic management. This paper focuses on the quality of macroeconomic policy formulation and implementation. The key independent variable, the CPIA Macroeconomic Management Rating, is drawn directly from this database and evaluated on a standardized scale ranging from 1 to 6.

In addition to the macroeconomic policy quality measure, several control variables are incorporated to account for potential omitted variable bias. Export volume data, used to capture a country's exposure to global trade shocks, is obtained from the World Bank's Development Indicators. Domestic credit to the private sector, expressed as a percentage of GDP, is included as a measure of financial sector development and liquidity conditions, drawn from the same source.

A composite indicator, the CPIA Average Rating, is created by calculating the mean of five critical CPIA sub-scores: economic management, financial sector development, social inclusion and equity, property rights and rule-based governance, and social protection. This custom variable captures broader institutional quality and policy coherence beyond macroeconomic management alone.

Together, the dataset integrates key themes such as governance quality, financial market development, trade openness, and social inclusion. These themes reflect the complex, multifaceted drivers of inflation and economic stability across countries. The diversity of the dataset ensures a comprehensive analysis of inflation outcomes across different institutional and economic environments, allowing for nuanced insights into the interplay between macroeconomic management and broader development structures.

Table 2:– Summary Statistics

Observation	**Obs**	**Min**	**Std. Dev**	**Min.**	**Max**
Inflation	856	37.43121	835.0617	-35.83668	24411.03

Macro	922	3.630694	.6785062	1	5.55
Export	866	242.8489	554.8185	10.24806	9589.121
CPIA_Avg	922	3.152354	.4864851	1.1	4.32
Credit	847	32.80088	25.57832	-41.18233	135.1

Empirical Model Specification

Regression Model and Theoretical Justification

We employ a fixed-effects linear model with the clustered panel data framework, operationalized through an ordinary least squares (OLS) strategy. The model to be estimated is articulated as follows

$$Inflation_{it} = \beta_0 + \beta_1 Macro_{it} + \beta_2 Export_{it} + \beta_3 Macro_{it} \times Export_{it} + \beta_4 CPI\ Avg_{it} + \beta_5 Credit_{it} + \alpha_i + \gamma_t + \epsilon_{it}$$

Where $Inflation_{it}$ represents the inflation in consumer prices for country i at time t. The model uses $Macro_{it}$ as its primary explanatory variable, and a proposed interaction term $Macro_{it} \times Export_{it}$, control variables $Export_{it}$, $Credit_{it}$,$CPI\ Avg_{it}$, and representing the other cluster control variables we use element fixed-effects α_i, time fixed-effects γ_t and an error term ϵ_{it}

We expect the sign of the coefficient β_1 to be negative. Effective macroeconomic management strengthens a country's ability to stabilize prices by anchoring inflation expectations, maintaining fiscal discipline, and mitigating demand-side pressures.

Countries with higher macroeconomic policy ratings are characterized by stronger institutional frameworks and more credible policy environments, both of which are associated with lower inflation volatility. As well, aligns with public expectation that strong macroeconomic policy can help curb the impacts of inflation.

Variable Selection Rationale

The primary explanatory variable of interest in this study is CPIA Macroeconomic Management Rating (Macro), a continuous measure of macroeconomic policy quality on a standardized 1–6 scale. Its inclusion in the model is based on both theoretical motivation and empirical performance. Across all estimated models, Macro consistently demonstrates strong statistical significance at the 1% level. For instance, in the preferred interaction model (Column 2 of Table 3), the coefficient on Macro is -4.644 ($p<0.01$), supporting the hypothesized negative relationship between improved macroeconomic management and lower inflation rates. Even in the absence of additional interaction terms or controls (Column 1), Macro remains highly significant with a coefficient of -3.222 ($p<0.01$), underscoring its centrality to the study's research question.

Export Volume (Export) is included both as an independent control and as part of an interaction term with Macro. The rationale stems from the potential influence of external trade shocks on domestic inflation dynamics. While Export alone is marginally significant in the final model ($p<0.1$), the interaction term (Macro × Export) achieves statistical significance at the 10% level (coefficient = 0.00792; $p<0.1$). This significance justifies its inclusion, allowing the analysis to capture how macroeconomic policy effectiveness may vary based on a country's degree of integration into global markets.

CPIA Average Rating (CPIA_Avg), a custom composite measure of broader institutional quality, is included to control for the potential confounding influence of a country's financial sector development, property rights protections, and social inclusion efforts. Although CPIA_Avg exhibits only marginal statistical significance ($p<0.1$) in the interaction model, its inclusion is critical. Omitting CPIA_Avg leads to substantial omitted variable bias, shifting the estimated effect of Macro significantly and reducing its interpretability. Its presence ensures that the estimated relationship between macroeconomic management and inflation is not conflated with broader governance quality effects.

Domestic Credit to Private Sector (Credit Financial), expressed as a percentage of GDP, is incorporated to account for liquidity conditions and credit market dynamics, which the literature identifies as important determinants of inflation. This variable demonstrates consistent statistical significance at the 1% level across all specifications (e.g., -0.0741; $p<0.01$ in Column 2), further validating its inclusion.

In evaluating potential omitted variables, we systematically tested several additional controls across multiple specifications. These included GDP per capita growth, real GDP levels, depth of credit information systems, electricity access rates (as a proxy for infrastructure quality), and levels of informal payments (as a proxy for corruption). Models including GDP growth and GDP constant dollars did not yield any meaningful change in the

Macro coefficient and were therefore excluded. Including **depth of credit** introduced slight changes but substantially reduced the number of observations, undermining estimation robustness. Similarly, including **electricity access** and **informal payments** did not produce statistically significant shifts in the key coefficients, nor did they meet the threshold of moving the estimated coefficient on Macro by more than one standard error. As such, these variables were ultimately omitted from the final model to preserve sample size and model parsimony.

Additionally, the possibility of nonlinearities was considered by estimating models with quadratic terms and log-linear transformations. However, none of these alternative specifications improved model fit, as assessed by adjusted R-squared values and statistical significance tests. Consequently, a linear specification with an interaction term between Macro and Export was selected as the most theoretically coherent and empirically robust model for this analysis.

Overall, the included variables have been carefully selected based on a combination of statistical significance, theoretical justification, and a thorough assessment of potential omitted variable bias. This ensures that the final model provides a rigorous and interpretable estimate of the relationship between macroeconomic policy quality and inflation outcomes across countries.

Table 3: - Estimates of Macroeconomic Rating: Effect on Change in Inflation across countries and time

VARIABLES	(1) inflation	(2) inflation	(3) inflation
Macro	-3.222***	-608.9	-4.644***
	(0.691)	(564.5)	(1.104)
Export	4.56e-06	-6.812	-0.0317*
	(0.000260)	(6.438)	(0.0163)
CPIA Avg	3.483**		3.284*
	(1.737)		(1.776)
Credit	-0.0773***		-0.0741***
	(0.0219)		(0.0215)
Macro x Export (Interaction Term)		1.703	0.00792*
		(1.610)	(0.00407)
Constant	10.51**	2,376	16.67**
	(5.149)	(2,169)	(6.451)
Observations	794	825	794
Number of country	75	75	75
Adjusted R-squared	0.039	0.022	0.044

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Results

The coefficient on the CPIA macroeconomic management rating is −4.644 ($p<0.01$), indicating that higher macroeconomic policy quality is significantly associated with lower inflation rates. Export volume alone has a small, marginally significant negative effect on inflation ($p=0.055$). Critically, the interaction term between CPIA macro rating and export volume is positive and marginally significant ($p=0.055$), implying that the deflationary impact of macroeconomic policy quality weakens as export volume increases.

Domestic credit to the private sector is also shown to significantly reduce inflation ($p<0.01$), emphasizing the importance of stable financial sector conditions. CPIA average rating, although only marginally significant ($p=0.068$), proves essential; without controlling for this broader measure of social, economic, and institutional quality, the relationship between macroeconomic management and inflation loses statistical significance, indicating negative Omitted Variable Bias. Thus, a nation's broader commitment to debt management, social inclusion, and property rights is crucial for macroeconomic policy to effectively curb inflation.

Overall, better macroeconomic management is strongly associated with lower inflation outcomes. However, this effect is moderated by export volumes: countries with larger export sectors experience a weaker inflation-reducing benefit from improvements in macroeconomic policy. Financial development, as proxied by domestic credit to the private sector, also plays a significant role in containing inflation. While broader institutional quality, captured by the CPIA Average Rating, must be controlled to accurately estimate the impact of macroeconomic management, its direct relationship with inflation is positive and marginally significant. This suggests that improvements in areas such as social inclusion and property rights, while important for overall governance, do not necessarily translate into lower inflation rates in the short to medium term.

Limitations and Future Research

While the models explain some of the variation in inflation outcomes (Adjusted R-squared approximately 5%), a substantial portion remains unexplained. Future research could enhance explanatory power by incorporating additional variables such as fiscal deficits, political stability indices, and more granular trade shock measures.

The dataset's temporal limitation (ending in 2016) suggests the need for updates, particularly given major economic disruptions in the late 2010s and early 2020s. Moreover, while CPIA Average Rating serves as a useful control, it introduces some multicollinearity with the macroeconomic management index. Future studies could refine these hypotheses or apply instrumental variable techniques to address potential biases.

Finally, moving beyond fixed effects to dynamic panel models or single-country time series analyses could provide richer insights into the causal mechanisms at play for specific nations.

Conclusion

This study finds that the quality of a nation's macroeconomic policy exerts a statistically significant influence on inflation outcomes, but that this relationship is conditioned by export activity levels. Specifically, while improvements in macroeconomic management ratings are associated with reductions in inflation, the marginal benefit of these improvements diminishes as export volumes increase.

Importantly, macroeconomic policy quality only emerges as a significant predictor of inflation once broader structural factors—including domestic financial sector stability, dedication to debt management, social inclusion, and property rights protections—are controlled for. Policymakers seeking to reduce inflation cannot rely solely on macroeconomic fine-tuning; rather, effective policy must be complemented by robust financial sectors, equitable social policies, and strong institutional frameworks.

Moreover, these findings suggest that informed voter assessments of a government's ability to manage inflation should extend beyond simple evaluations of macroeconomic policies. Given the interconnectedness of inflation outcomes with broader social and institutional structures — including debt management strategies and commitments to social inclusion — citizens and policymakers alike should consider the quality of governance in these areas when forming expectations about future inflation dynamics. A strong macroeconomic framework, while necessary, is insufficient on its own without accompanying institutional strength and inclusive development policies.

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